

# Biochemistry:

## Physical Units:

limit of a light microscope:  $2000 \text{ \AA}$  or  $0.2 \mu\text{m}$

milli ms: enzymatic reactions (酶促反应)

$\text{ATP} \rightarrow \text{ADP} + \text{P}_i$  :  $7.3 \text{ kcal/mol} = 30.5 \text{ kJ/mol}$

biomolecules 生物分子

## Chemical Bonds:

### • covalent bond 共价键

valence electron 价电子

single/double/triple bond 单/双/三键

polar/non-polar bond 极性/非极性键

polarity 极性

dipole 偶极 (a molecular or covalent bond that has a separation of charges)  
bonds between phosphate groups in ATP:  
磷酸基团

electronegative charges exerting a repelling force

### • noncovalent bond 非共价键

ionic bond 离子键

Van der Waals force 范德华力

(characteristics: significantly weaker, no saturation  
or directional character)  
方向性 饱和性

h

hydrogen bond 氢键

acceptor (电子)受体 donor (电子)供体

hydrophobic interaction 疏水作用

decrease the interfacial area between the  
hydrophobic species and water (thermodynamic <sup>热力学稳定性</sup> stability)  
in: lipid <sup>脂质膜</sup> membrane, protein <sup>蛋白质折叠</sup> folding

## Major Biomolecules:

protein 蛋白质

nuclear acid 核酸

lipid 脂质

carbohydrate 糖类

cell membrane 细胞膜 glycosylation 糖基化

glycoprotein 糖蛋白 glycolipid 糖脂

carbohydrate functions: provide/store energy,

build <sup>大分子</sup> macromolecules, spare protein and fat

glycogen 糖原 (muscle <sup>肌肉</sup> and liver <sup>肝脏</sup>)

## The Basic Structure of Protein:

- amino acid 氨基酸 (20-21 types)

amine 氨基 carboxyl 羧基

D and L isomers D/L 异构体

Almost all biological amino acids are in the L-configuration

determine: look at  $\alpha$ -carbon (hydrogen atom behind it),  $\text{COOR} \rightarrow \text{R} \rightarrow \text{NH}_2 / \text{CON}$ , counterclockwise - L, clockwise - D

| Non-Polar  | Carboxyl      | Amine     | Aromatic      | Hydroxyl  | Other          |
|------------|---------------|-----------|---------------|-----------|----------------|
| Alanine    | Aspartic Acid | Arginine  | Phenylalanine | Serine    | Asparagine     |
| Glycine    | Glutamic Acid | Histidine | Tryptophan    | Threonine | Cysteine       |
| Isoleucine |               | Lysine    | Tyrosine      | Tyrosine  | Glutamine      |
| Leucine    |               |           |               |           | Selenocysteine |
| Methionine |               |           |               |           |                |
| Proline    |               |           |               |           |                |
| Valine     |               |           |               |           |                |

| Name          | Charged R-group               | Approximate R-group pKa |
|---------------|-------------------------------|-------------------------|
| Aspartic Acid | -COO <sup>-</sup>             | 4.1                     |
| Glutamic Acid | -COO <sup>-</sup>             | 4.1                     |
| Lysine        | -NH <sub>3</sub> <sup>+</sup> | 10.5                    |
| Arginine      | =NH <sub>2</sub> <sup>+</sup> | 12.5                    |
| Histidine     | =-NH <sup>+</sup>             | 6.0                     |
| Cysteine      | -S <sup>-</sup>               | 8.2                     |
| Tyrosine      | -O <sup>-</sup>               | 10.5                    |

aromatic R-group 芳香R基团

nonpolar R-group 非极性R基团

R-group carboxyl R基上羧基

ionization 电离

hydroxyl R-group 羟基R基团

Sulfhydryl R-group 巯基R基团

carboxamide R-group 羧酰胺R基团

ionizable amine R-group 可电离胺R基团

peptide bond 肽键 (N-terminus, C-terminus)

dipeptide 二肽 polypeptide 多肽

## • protein structure

primary: <sup>序列</sup>sequence of amino acid

<sup>合成</sup>Synthesis: begins at amino end  $\rightarrow$  terminates at carboxyl end  
 $N \rightarrow C$

secondary: interaction of amino acids close in primary sequence

$\alpha$ -helix <sup>螺旋</sup>,  $\beta$ -sheet (hydrogen bonds, carbonyl O and amino H)  
between atoms of the backbone <sup>主链</sup> (not involve R group)

tertiary: interaction of amino acids distantly located

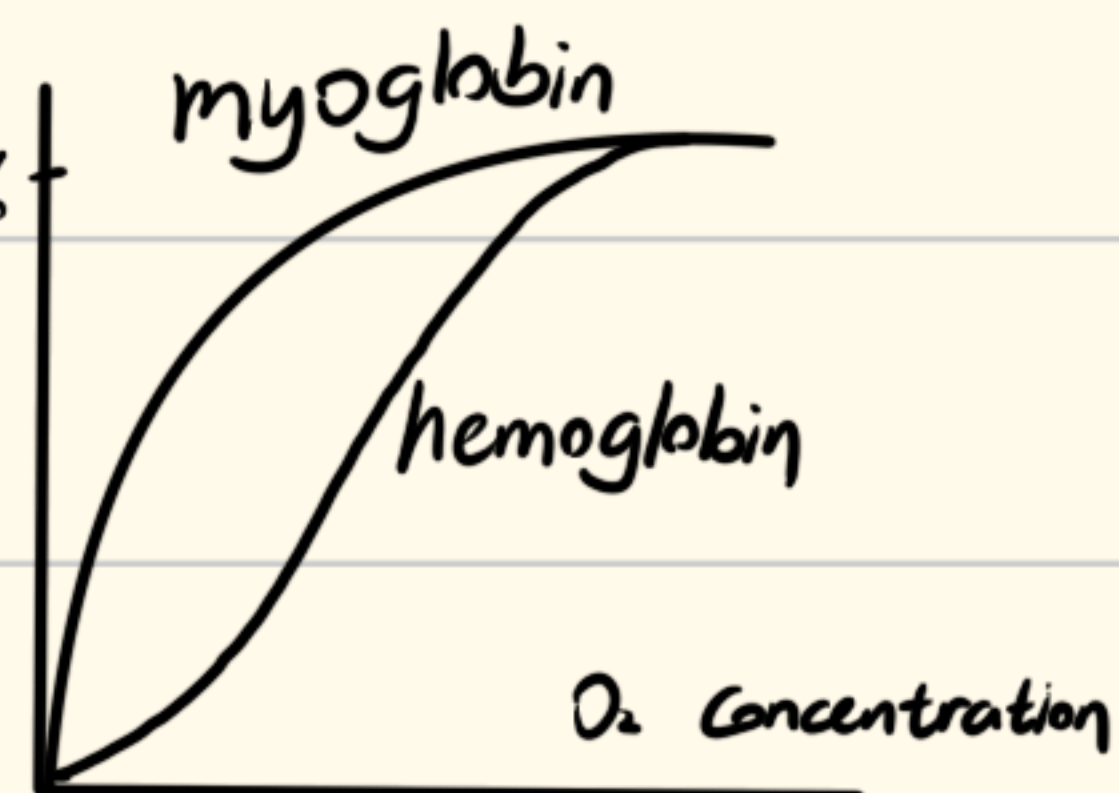
between the side chain <sup>侧链</sup> and the backbone groups <sup>主链</sup>

quaternary: interaction of protein subunits <sup>蛋白质亚基</sup>

held by hydrogen bonds and van der Waals forces

hemoglobin 血红蛋白 4 subunits 100%

myoglobin 肌红蛋白 1 subunit



cooperativity of hemoglobin

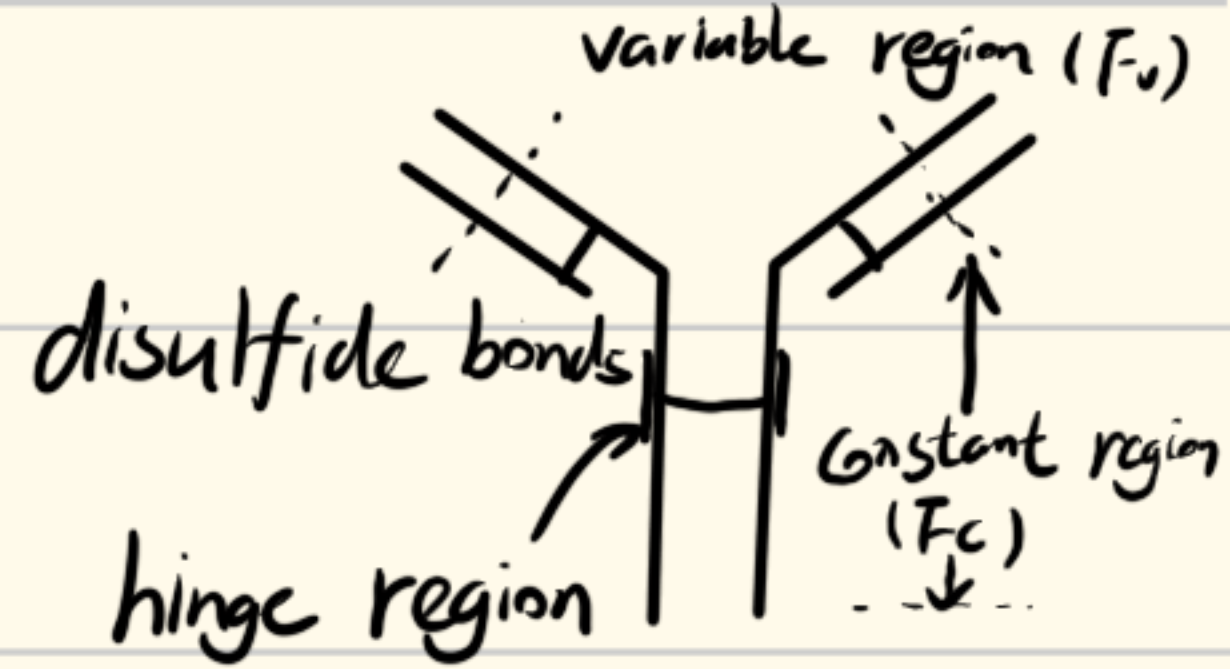
Bohr effect:  $pH \downarrow / CO_2 \uparrow \Rightarrow$  less oxygen binding

antibody 抗体

light chain 轻链(L) heavy chain 重链(H)

constant region 恒定区

variable region 可变区



bound: disulfide bonds, hydrogen bonds, and hydrophobic interactions.

5 H chains and 2 L chains.

IgG, IgM, IgA, IgD, IgE

hinge region 铰链区

Ig fold (immunoglobulin fold 免疫球蛋白折叠)

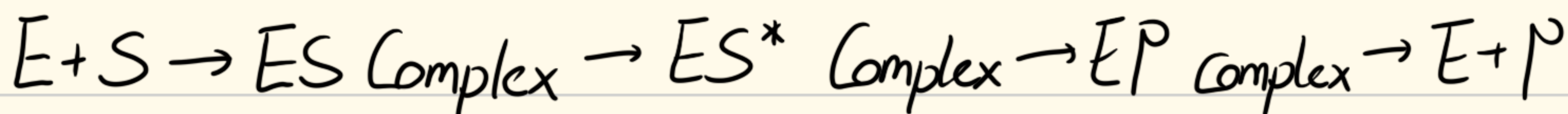
## Enzyme:

enzymatic catalysis

酶催化

enzymatic reaction

酶促反应



reversible 可逆的

induced fit 诱导契合

equilibrium 平衡态

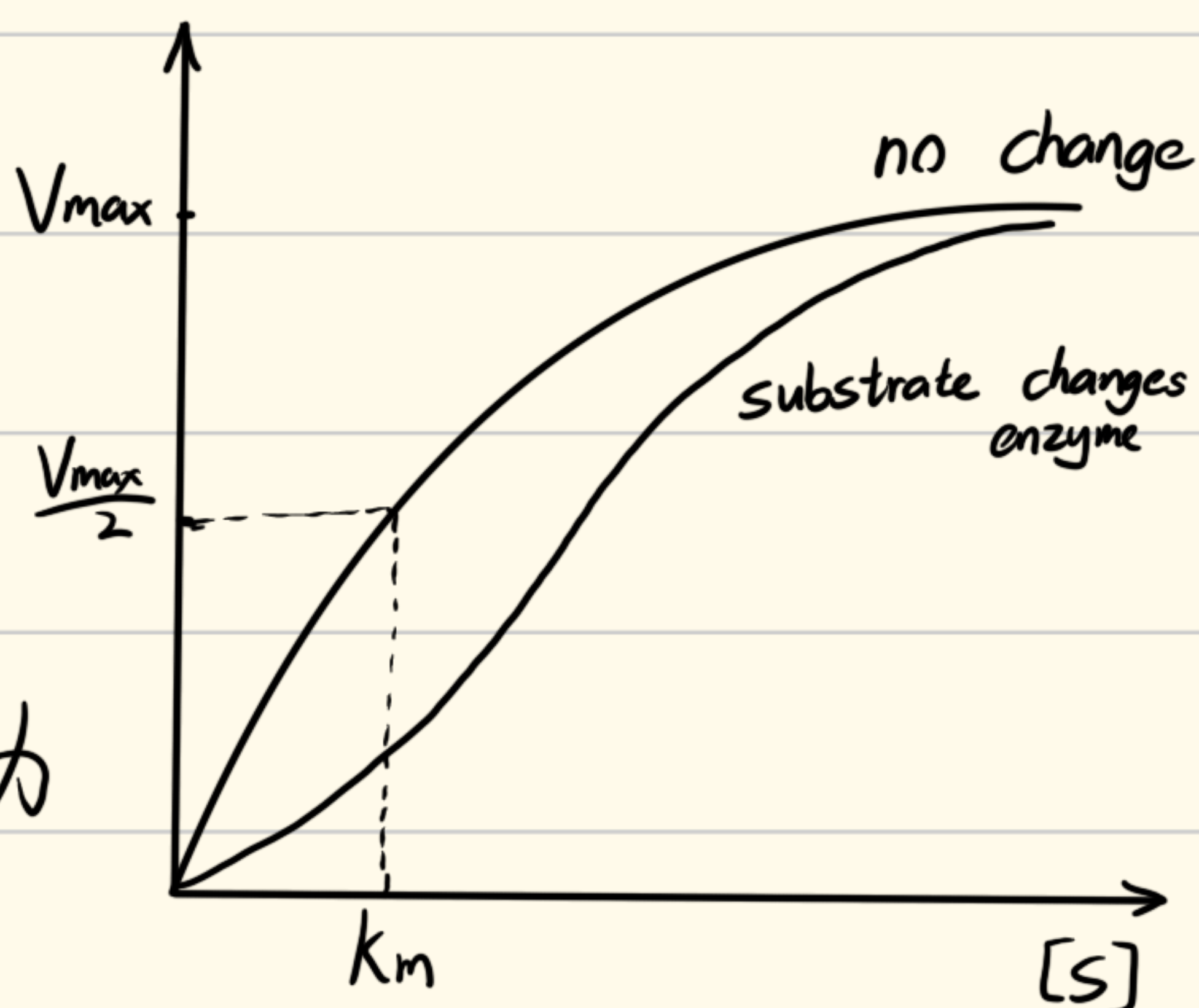
activation energy 活化能

substrate 底物

enzyme's affinity 酶亲和力

$K_m$  value inversely related

to affinity.



enzymatic inhibition 酶抑制

competitive inhibitor 竞争性抑制剂

noncompetitive inhibitor 非竞争性抑制剂

# Cell:

## Basic Structure:

prokaryote 原核细胞

eukaryote 真核细胞

ER 内质网 golgi 高尔基体

mitochondrion 线粒体 cytoplasm 细胞质

## Protein Sorting:

endomembrane system 内膜系统

cytosolic protein 胞质蛋白

signaling molecule 信号分子

catabolic enzyme 代谢酶

posttranslational sorting 翻译后分选

nuclear protein 核蛋白

mitochondrial protein 线粒体蛋白

cotranslational sorting 翻译中分选

PM-membrane protein 膜蛋白

secretory factor 分泌因子

Posttranslational sorting: chloroplast 叶绿体,

mitochondrion, nucleus, peroxisome 过氧化物酶体

NLS (nuclear localization signal),

MTS (mitochondrial targeting signal)

## Cotranslational sorting:

ER signal sequence 内质网信号肽

SRP 信号识别颗粒 (RNA+protein)

secretory vesicle 分泌囊泡

lysosome 溶酶体

## Vesicle Trafficking:

vesicle-mediated transport 囊泡介导的运输

secretion/exocytosis 分泌

endocytosis 内吞

intracellular transport 细胞内运输

endoplasmic reticulum 内质网

rough ER 粗面内质网 smooth ER 光面内质网

exocytosis 胞吐作用

passive transportation 被动运输

simple diffusion 简单扩散

facilitated diffusion 协助扩散

active transportation 主动运输

pinocytosis 胞饮

macropinocytosis 巨胞饮

receptor-mediated endocytosis:  
clathrin 网格蛋白

phagocytosis 吞噬作用

microtubule-mediated 微管介导

## Cytoskeleton-mediated Transportation:

cytoskeleton 细胞骨架

microtubule 微管

actin filament 微丝

intermediate filament 中间纤维

(microtubule) T form: grow, D form: shrink

kinesin (plus end), dynein (minus end), dimer

motor protein 马达蛋白

## Protein Quality Control:

protein folding:

co-translational protein folding

molecular chaperone 分子伴侣

heat-shock protein 热激蛋白 (hsp)

hsp 60, hsp 70, hsp 90

protein misfolding:

indication: exposed hydrophobic patch 疏水区

a continued presence of a chaperone

ubiquitin-proteasome 泛素-蛋白酶体

recognized by exposed hydrophobic patch

K48 ubiquitin chain 降解信号

E1 泛素酶 E2 整合酶 E3 连接酶

protein degradation 蛋白降解

ubiquitin hydrolase 泛素水解酶 (recycle)

unfold 去折叠

protein folding marks in the ER:

glycosylated 糖基化

mannosidase 甘露糖酶 (export/degrade)

autophagy 自噬

autophagosome 自噬小体

non-selective autophagy 非选择性自噬

selective autophagy 选择性自噬 (mitophagy/线粒体自噬)

ubiquitin chain, ATG protein LC3 自噬蛋白, lysosome

## The Cell Circle:

duplication 复制, segregation 分配, division 分裂